

5. Design of slabs

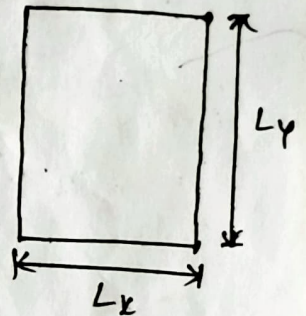
Slabs are plain structural members whose thickness is small as compared to its length and breadth.

Slabs are more frequently used as roof covering and floors in various shapes such as square, rectangle, circular, triangle etc.

Types of slabs

Depending upon the ratio of longer span to shorter span (L_y/L_x), the slabs are classified into two types.

1. One way slab
2. Two way slab



One way slab

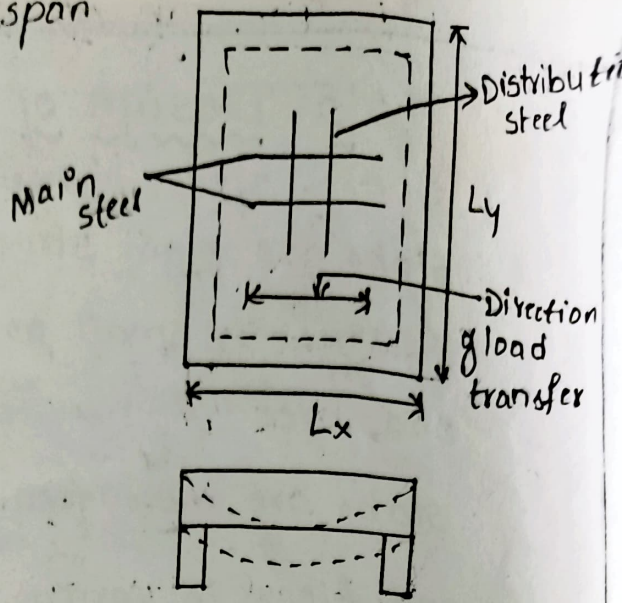
Slabs which are supported ^{on two edges} ~~on all four edges~~ and the ratio of longer span to the shorter span i.e. $\frac{L_y}{L_x} > 2$ are known as one way

slab:

* one way slab bends in one direction i.e. along the shorter span, hence it needs main reinforcement in one direction only i.e. the shorter span, to resist the one way bending.

And here the minimum reinforcement is known as distribution steel which is provided

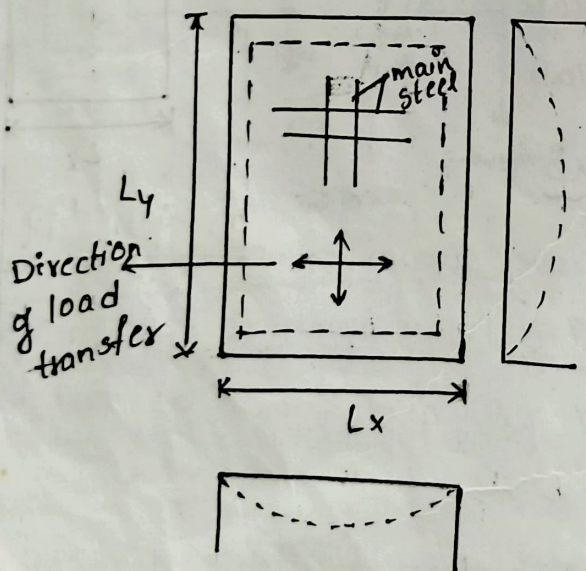
along the longer span



Two way slabs

When the slabs are supported on all the four edges and the ratio of longer span to the shorter span i.e. $\frac{L_y}{L_x} \leq 2$ then it is known as known as two way slabs.

Two way slabs bend along the two spans



General design procedure for one-way slab.

1. Effective span

The effective span of a simply supported span

is taken as

- (i) clear span + effective depth (or)
- (ii) centre to centre distance between supports
which ever is less

2. Limiting stiffness: (Pg: 37) (maximum effect)

The stiffness of slabs is governed by the span to depth ratio. As per clause 23.2 of IS 456-2000 for spans not exceeding 10m, the span to depth ratio should not exceed the limit given below

Cantilevers	-	7
Simply supported	-	20
Continuous	-	28

3. Minimum reinforcement

The reinforcement in either direction of span shall not be less than 0.15% of gross cross-sectional area. If mild steel is used however, this value is reduced to 0.12% where high yield strength deformed bars are used (clause 23.2.1)

4. Maximum diameter of bars:

The diameter of the bar shall not exceed $\frac{1}{8}$ of the total ~~deflection~~ ^{thickness} of slab. (clause 20.3.1)

5. Spacing of main reinforcement

Spacing of main reinforcement in slabs shall not be more than 3 times the effective

depth of the slab (or) 300mm whichever ever is less (clause 26.3.3)

6. Distribution reinforcement

The area of distribution reinforcement shall not be less than 0.15% of gross cross-sectional area if plain bars are used and 0.12% if high yield strength deformed bars are used

The spacing of distribution reinforcement shall not be more than 5 times the effective depth of the slab (or) 450mm whichever ever is less.

7. cover to reinforcement

Reinforcement shall have concrete cover of thickness as follows:

- (i) at the end of reinforcement bars not less than 25mm nor less than twice the diameter of that bar
- (ii) The bottom cover of reinforcement shall not be less than 20mm nor less than diameter of bar

Design procedure of one way slab

1. Assume the suitable depth based on stiffness condition.

span by depth ratio can be selected in the

range of 25 to 30 for simply supported span.

2. calculate the effective span

3. calculate loads, bending moment and shear force.

4. calculate minimum depth required, if it is not satisfied then modify the depth of the beam.

5. calculate (tension reinforcement) Area of steel

6. find the spacing of bars

$$S = \frac{A_{st}}{A_{st}} \times 1000$$

where A_{st} = area of bar

A_{st} = total area of steel required

spacing should not be more than $3d$ (or) 300mm, whichever is less

d = effective depth.

7. Distribution steel $A_{t(D.S)} = \frac{0.15\% \text{ bD}}{\text{Fe 500}}$

The area of distribution reinforcement shall not be less than 0.15% of gross cross-section area if plain bars are used and 0.12% of high yield strength deformed bars are used Fe 500

The spacing of distribution reinforcement shall not be more than 5 times the effective depth of the slab (or) 450mm whichever is less

8. Design of shear reinforcement

9. check for deflection

1. A simply supported R.C.C slab has to be provided for the roof of a room of clear dimensions of $3 \text{ m} \times 8 \text{ m}$. Width of the supported wall is 300 mm , the weight of weathering course over the slab is 1 kN/m^2 . Take live load as 2 kN/m^2 . Design the slab use M20 and Fe415

Sol

Given

$$\text{live load} = 2 \text{ kN/m}^2$$

$$\text{floor finish} = 1 \text{ kN/m}^2$$

$$f_{ck} = 20 \text{ N/mm}^2$$

$$l = 3 \text{ m}$$

$$f_y = 415 \text{ N/mm}^2$$

Step 1:

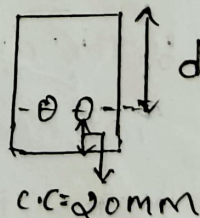
$$\frac{l_y}{l_x} = \frac{8}{3} = 2.67 > 2$$

Hence it should be designed as one way slab.

Step 2: Depth

$$\text{Assuming depth} = \frac{\text{span}}{25} = \frac{3000}{25} = 120 \text{ mm}$$

$$\begin{aligned} \text{(clear cover)} &= 120 \text{ mm} \\ \text{(diameter of bar)} &= 10 \text{ mm} \\ D &= 120 + 20 + \frac{10}{2} = 145 \text{ mm} \end{aligned}$$



Step 3 (Adopting clear cover of 20 mm for 10 mm diameter of bars)

$$\text{Effective span} = \text{clear span} + d = 3 + 0.14 = 3.14 \text{ m}$$